UIT2521 – Information Theory and Applications

UNIT II INFORMATION THEORY FUNDAMENTALS

Tutorial – V

Joint and Conditional Entropy

Date & Time: 25.07.2025 (Friday, 1st Hour)

1. A discrete source transmits message x_1, x_2, and x_3 with the probabilities 0.3, 0.4, 0.3. The source is connected to the channel as given in Fig. 1. Calculate all the entropies.

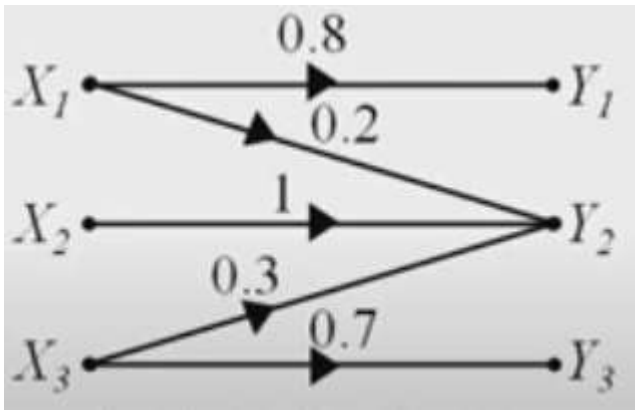

Fig. 1. Conditional Probability Matrix

Step 1. Source probabilities

We are given:

$$P(X_1) = 0.3, \quad P(X_2) = 0.4, \quad P(X_3) = 0.3.$$

Step 2. Channel transition probabilities (from the figure)

- From X_1:
  $P(Y_1|X_1) = 0.8,\ P(Y_2|X_1) = 0.2,\ P(Y_3|X_1) = 0.$
- From X_2:
  $P(Y_2|X_2) = 1.0$ (deterministic).
- From X_3:
  $P(Y_2|X_3) = 0.3,\ P(Y_3|X_3) = 0.7,\ P(Y_1|X_3) = 0.$

Step 3. Joint distribution $P(X, Y)$

$$P(X_i, Y_j) = P(X_i)P(Y_j|X_i).$$

Compute:

- For X_1:
  $P(X_1, Y_1) = 0.3 \times 0.8 = 0.24.$
  $P(X_1, Y_2) = 0.3 \times 0.2 = 0.06.$
  $P(X_1, Y_3) = 0.3 \times 0 = 0.$
- For X_2:
  $P(X_2, Y_1) = 0,$
  $P(X_2, Y_2) = 0.4 \times 1 = 0.4,$
  $P(X_2, Y_3) = 0.$
- For X_3:
  $P(X_3, Y_1) = 0,$
  $P(X_3, Y_2) = 0.3 \times 0.3 = 0.09,$
  $P(X_3, Y_3) = 0.3 \times 0.7 = 0.21.$

✅ Check total: $0.24 + 0.06 + 0.4 + 0.09 + 0.21 = 1.0.$

Step 4. Output distribution $P(Y)$

- $P(Y_1) = 0.24.$
- $P(Y_2) = 0.06 + 0.4 + 0.09 = 0.55.$
- $P(Y_3) = 0.21.$

Step 5. Entropies

(a) Source entropy $H(X)$

$$H(X) = -\sum_i P(X_i) \log_2 P(X_i).$$

$$H(X) = -[0.3 \log_2 0.3 + 0.4 \log_2 0.4 + 0.3 \log_2 0.3].$$

- $0.3 \log_2 0.3 \approx 0.3 \times (-1.737) = -0.521.$ (two such terms → -1.042).
- $0.4 \log_2 0.4 \approx 0.4 \times (-1.322) = -0.529.$

So,

$$H(X) \approx 1.571 \text{ bits.}$$

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(b) Output entropy $H(Y)$

$$H(Y) = -[0.24 \log_2 0.24 + 0.55 \log_2 0.55 + 0.21 \log_2 0.21].$$

- $0.24 \log_2 0.24 \approx 0.24 \times (-2.059) = -0.494.$
- $0.55 \log_2 0.55 \approx 0.55 \times (-0.862) = -0.474.$
- $0.21 \log_2 0.21 \approx 0.21 \times (-2.248) = -0.472.$

Sum = -1.440.

$$H(Y) \approx 1.440 \text{ bits.}$$

(c) Joint entropy $H(X, Y)$

$$H(X, Y) = -\sum_{i,j} P(X_i, Y_j) \log_2 P(X_i, Y_j).$$

Nonzero entries:

- $0.24 \log_2 0.24 \approx -0.494.$
- $0.06 \log_2 0.06 \approx -0.244.$
- $0.4 \log_2 0.4 \approx -0.529.$
- $0.09 \log_2 0.09 \approx -0.314.$
- $0.21 \log_2 0.21 \approx -0.472.$

Sum = 2.053 bits.

$$H(X, Y) \approx 2.053 \text{ bits.}$$

(d) Conditional entropy $H(Y|X)$

By definition:

$$H(Y|X) = H(X, Y) - H(X).$$

$$H(Y|X) \approx 2.053 - 1.571 = 0.482 \text{ bits.}$$

(e) Conditional entropy $H(X|Y)$

$$H(X|Y) = H(X, Y) - H(Y).$$

$$H(X|Y) \approx 2.053 - 1.440 = 0.613 \text{ bits.}$$

(f) Mutual information $I(X; Y)$

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

$$I(X; Y) \approx 1.571 - 0.613 = 0.958 \text{ bits.}$$

✅ **Final Results (all entropies)**

- Source entropy: $H(X) \approx 1.571$ bits
- Output entropy: $H(Y) \approx 1.440$ bits
- Joint entropy: $H(X, Y) \approx 2.053$ bits
- Conditional entropies:
  - $H(Y|X) \approx 0.482$ bits
  - $H(X|Y) \approx 0.613$ bits
- Mutual information: $I(X; Y) \approx 0.958$ bits